

Deeksha Reddy Patlolla

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Education

Master of Science in Computer Science

Denver, CO

University of Colorado Denver (GPA: 3.9/4.0)

Aug 2024 – May 2026

- **Relevant Coursework:** Machine Learning, Big Data Science, Computer Vision, Advanced Computer Architecture, Cyber/Infrastructure Defense, Algorithms, Social Networks & Informatics, Theory of Automata.

Experience

University of Colorado, Denver

Denver, CO

Data Engineer

Mar 2025 – May 2026

- **Data Modeling & Schema Design:** Designed a normalized relational schema across 8 PostgreSQL tables with composite indexes and materialized views (refreshed concurrently without downtime), reducing average dashboard query time from 3.2s to 1.9s – a **40% improvement**.
- **API Development:** Built a **FastAPI** backend exposing both REST and **GraphQL** endpoints using Strawberry and SQLAlchemy async ORM; implemented query batching to eliminate redundant database calls, cutting per-request round-trips by **60%**.
- **Large-Scale ETL Pipelines:** Processed **10M+ row** datasets on **AWS EMR** using **PySpark**, applying window functions for feature engineering, optimized join strategies, and predicate pushdown – reducing pipeline runtime from 22 to 9 minutes (**59% faster**).
- **Data Warehouse & Transformation:** Authored layered **dbt** transformation models on **Snowflake** with incremental loading and automated schema contract tests, achieving **zero data quality violations** in production since deployment.
- **ML Model Integration:** Trained a **Scikit-learn GradientBoostingClassifier** (`n_estimators=300`, `max_depth=5`, 5-fold CV) on features engineered from 3 PostgreSQL tables; serialized with **joblib** and deployed as a **FastAPI** inference endpoint serving predictions at **<200ms p99 latency**.
- **Pipeline Orchestration & Quality:** Built a **Great Expectations** data validation suite integrated into **Apache Airflow** DAGs with automated failure alerts; used dynamic task mapping to eliminate manual pipeline steps, reducing intervention time by **50%**.

Infor Pvt Ltd

Hyderabad, India

Software Engineer

May 2023 – May 2024

- **Backend Engineering:** Built and maintained **Java/Spring Boot** RESTful microservices for enterprise ERP modules, applying service-oriented architecture patterns with **Hibernate/JPA** ORM – deployed to production serving **1,000+ business users** reliably.
- **CI/CD & Containerization:** Containerized services with **Docker** and deployed to **Kubernetes**; set up automated **CI/CD pipelines** with Jenkins and GitHub Actions, cutting release cycle time while maintaining high test coverage via **JUnit** and **Mockito**.
- **Database Performance:** Identified and resolved slow query bottlenecks in **MySQL** and **PostgreSQL** by rewriting queries, adding targeted indexes, and optimizing stored procedures – improving response times across high-traffic service endpoints.

Technical Skills

Languages & Systems: Python, Java, SQL, TypeScript, JavaScript, C/C++, Bash.

Data Engineering: PySpark, Apache Spark, dbt, Snowflake, Apache Airflow, Kafka, ETL/ELT, Parquet, Pandas, NumPy, Great Expectations.

Backend & APIs: FastAPI, GraphQL (Strawberry), REST APIs, SQLAlchemy 2.0 async, Spring Boot, Pydantic, gRPC.

AI & ML: Scikit-learn, PyTorch, TensorFlow/Keras, LangChain, LangGraph, RAG, pgvector, Hugging Face Transformers, SHAP, NLP.

Cloud & Infrastructure: AWS (EC2, S3, RDS, EMR, Lambda, SQS), Azure (AKS, Monitor, App Insights, Key Vault), Docker, Kubernetes, CI/CD, GitHub Actions, Jenkins.

Databases & Testing: PostgreSQL, MySQL, MongoDB, Redis, Oracle, pytest, pytest-asyncio, JUnit, Mockito, httpx, TDD.

AI-Assisted Development: GitHub Copilot, Claude AI, Cursor, ChatGPT, Prompt Engineering.

Projects

CivicLens – Policy Simulation Platform | *LangGraph, LangChain, Python*

Apr 2026

- **Agentic Pipeline:** Built a data ingestion layer pulling live **US Census/FRED API** data into **ChromaDB** (sentence-transformer embeddings), orchestrating 3 parallel AI agents via a **LangGraph StateGraph** with `asyncio.gather()` fan-out, cutting end-to-end latency from ~45s to ~15s.

Heterogeneity-Aware Multimodal Learning for Disease Detection | *PyTorch, SHAP*

Dec 2025

- **Transformer Fusion:** Built a **PyTorch** pipeline fusing imaging, genomic, and clinical tabular data via **gated cross-attention** transformer fusion with **SHAP** interpretability, improving **ROC-AUC by 11%** over early-fusion baseline.

Resume AI Intelligence Platform | *LangChain, FastAPI, pgvector*

Nov 2025

- **RAG Pipeline:** Chunked documents (512 tokens, 64-token overlap) into **pgvector** with **HNSW** index for sub-50ms ANN retrieval; served via **FastAPI** with **ConversationSummaryBufferMemory**, cutting GPT-4 token cost by **28%**.
- **Multi-Agent Orchestration:** Built a **LangChain** multi-agent system with tool-calling and structured output parsing; implemented async request handling and rate-limit retry logic for production-grade **OpenAI API** reliability.

Autonomous Cybersecurity Defender | *FastAPI, GPT-4o, Python*

Sep 2025

- **Real-Time Inference:** Extracted 20+ network traffic features from raw PCAP data and applied **StandardScaler** pre-filtering to reduce API calls by **35%**; classified intrusions on **CICIDS2017** using **GPT-4o (94% accuracy)**, served via async **FastAPI** at ~2.5s latency.